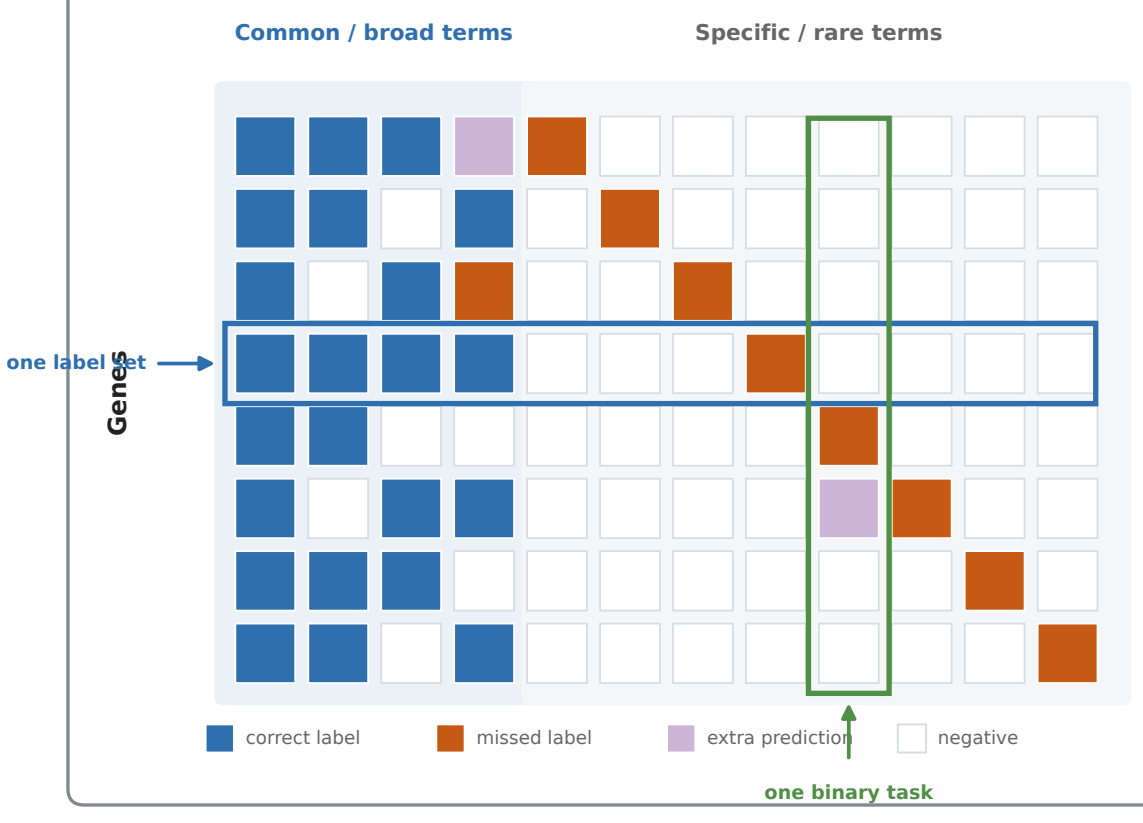
Why gene-level CAFA Fmax exceeds median GO-term Fmax

The metrics summarize the same predictions along different axes and weight common versus rare labels differently

A One prediction matrix, two evaluation directions

schematic



B Gene-level CAFA Fmax

For each gene

compute $P_i(t)$ and $R_i(t)$ over its GO-label set

average across genes → $\bar{P}(t)$ and $\bar{R}(t)$

$$F(t) = 2\bar{P}(t)\bar{R}(t)/(\bar{P}(t) + \bar{R}(t))$$

max over one global threshold

0.525

P 0.537 • R 0.514
t = 0.584

C GO-term median

3,381 TERMS

For each GO term

count TP_j, FP_j, and FN_j across genes

compute $F1_j$ → median across terms

Each term has equal weight—
rare and common alike

one global t

0.071

median F1

best t / term

0.220

median Fmax

WHY THE GAP? Common true-path ancestors improve many gene label sets + Numerous rare terms remain difficult → **stronger gene-level than median term-level score**